



POWER
PLATFORM

COURSE SLIDES · WSQ

Applications Integration with Power Apps and Power Automate

WSQ Course Code: TGS-2022015539

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE

Microsoft Power Platform, application integration, low-code development and business process automation.

DELIVERS

WSQ courses on Power Apps, Power Automate, Power BI and applications integration.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name, organisation and role.
- The systems you work with day to day — Excel, SharePoint, Outlook, a line-of-business app?
- One manual, repetitive process in your work that you would most like to automate.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

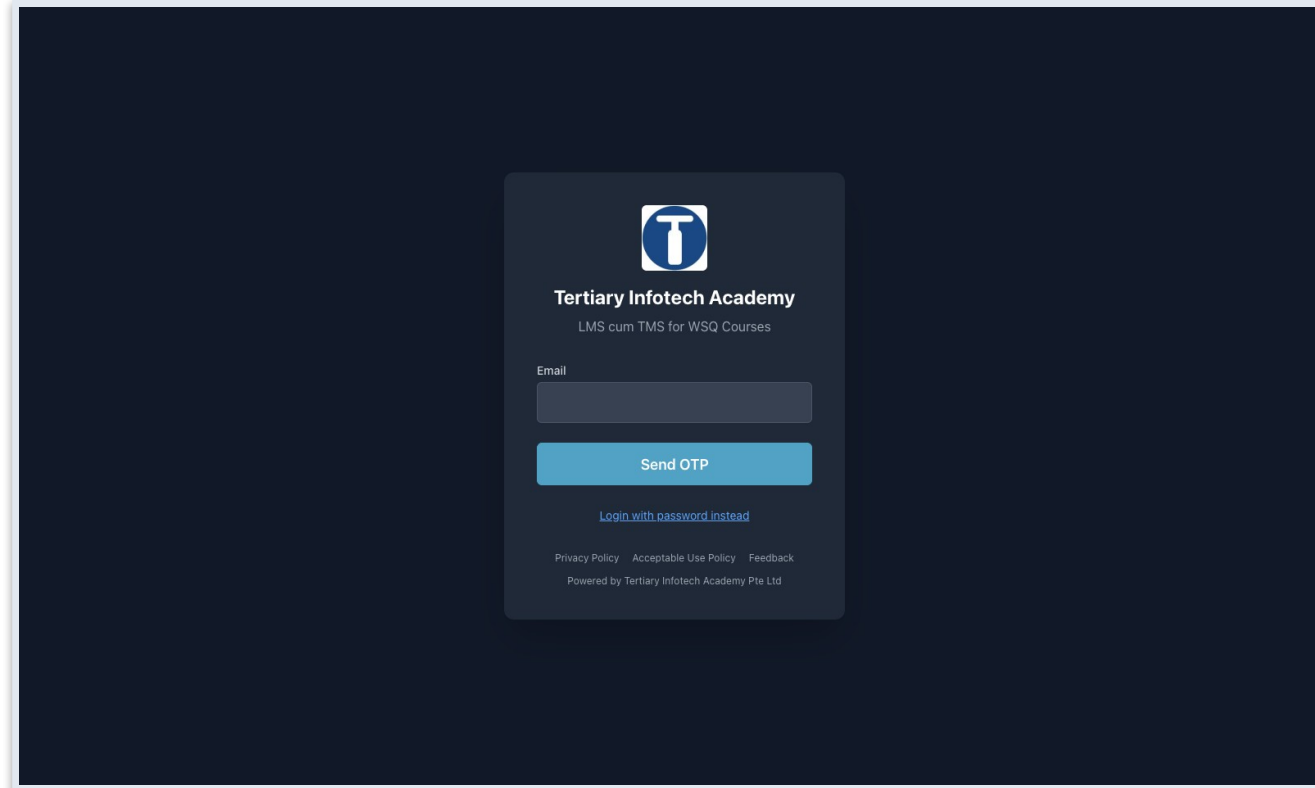
5

Be punctual; return from breaks on time.

6

75% attendance is required.

Download Your Course Material



WHAT TO LOOK AT

Log in with your registration e-mail, open this course, and download the Trainer Slides, Learner Guide and Lesson Plan. You may use them during the open-book assessment.

Skills Framework Alignment

1

TSC Title

Applications Integration

2

TSC Code

ICT-DIT-3003-1.1 · Proficiency Level 3

3

K1 / K2

Types of middleware and their features; proper usage of middleware.

4

K3 / K5

Platforms on which applications run; functions of Application Programming Interfaces.

5

K4

Potential technical, compatibility or performance issues in application integration.

6

A1 – A8

Identify opportunities, scan feasibility, integrate, support APIs, test, verify, highlight issues and mitigate them.

SCHEDULE

Lesson Plan — 2 Days, 8 hours/day

Day 1

- Day 1 — Opportunities, feasibility and building automation with Power Automate
 - Digital Attendance (AM) · Introductions · Learning Outcomes
 - Topic 1: Opportunities for Using Power Platform Apps (Labs 1–2)
 - Lunch Break · Digital Attendance (PM)
 - Topic 2: Power Automate (Labs 3–6)

Day 2

- Day 2 — Building canvas apps, integrating them with flows, and assessment
 - Topic 3: Power Apps (Labs 7–10)
 - Lunch Break · Digital Attendance (PM)
 - Topic 4: Integrate Power Apps and Power Automate (Labs 11–14)
 - TRAQOM Survey · Digital Attendance (Assessment)
 - Final Assessment (WA 1 hr + PP 1.5 hrs)

Learning Outcomes

1

LO1 · Identify opportunities

Identify opportunities to use Power Platform apps and perform a feasibility scan for potential application integration.

2

LO2 · Utilise Power Automate

Utilise and test Power Automate flows for business automation.

3

LO3 · Create Power Apps

Create Power Apps canvas apps and verify their functionalities.

4

LO4 · Improve with flows

Improve Power Apps with Power Automate flows, and resolve integration issues.

Course Outline

1

Topic 1 — Opportunities for Using Power Platform Apps

The integration landscape · middleware types · the Power Platform family · the feasibility scan · connectors.

2

Topic 2 — Power Automate

Cloud flow types · triggers and actions · dynamic content · conditions and branching · approvals · testing.

3

Topic 3 — Power Apps

Canvas vs model-driven · data sources · Power Fx · custom connectors and APIs · publishing and sharing.

4

Topic 4 — Integrate Power Apps and Power Automate

Calling a flow from an app · returning data · end-to-end approvals · troubleshooting and delegation.

BEFORE YOU START

Briefing for Assessment

1

Do · Clear your desk

Place phones and other materials under the table or on the floor.

2

Don't · No recording

No photos or recording of assessment scripts.

3

Don't · No discussion

Work individually — no discussion during the assessment.

4

Do · Black or blue pen

Use a black or blue pen for hard-copy assessments.

5

Don't · No correction fluid

No liquid paper or correction tape may be used.

6

Do · Stop on time

Scripts are collected when time is up.

Assessment

1

Written Assessment (WA)

Short-Answer Questions (SAQ) · 1 hour · open book.
Five questions covering K1–K5.

2

Practical Performance (PP)

Hands-on scenario tasks · 1.5 hours · open book. Four tasks covering A1–A8.

3

Open book

You may use the course slides, the Learner Guide and approved materials only.

4

Eligibility

A minimum of 75% attendance is required to be eligible for assessment and funding.

5

Result

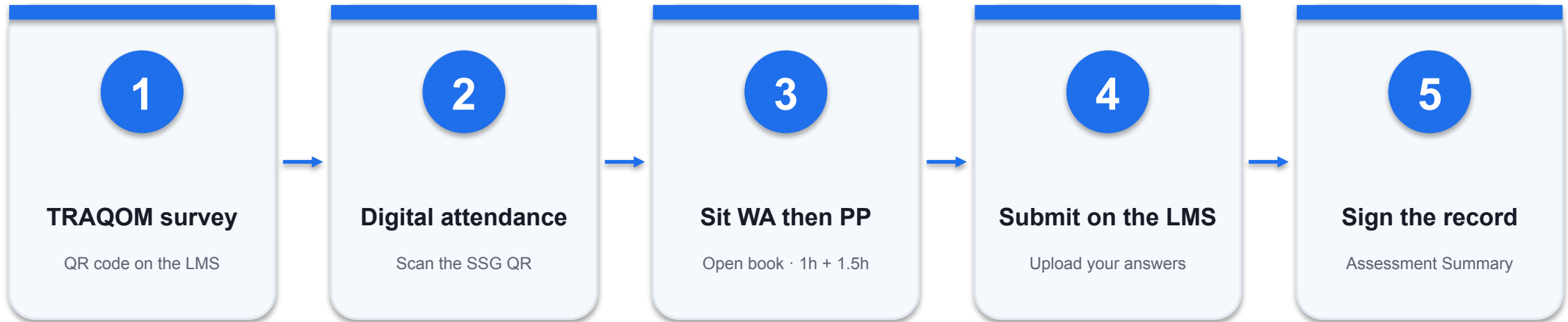
You are assessed as Competent (C) or Not Yet Competent (NYC) on each instrument.

6

Appeals

An appeal process is available if you wish to contest an assessment outcome.

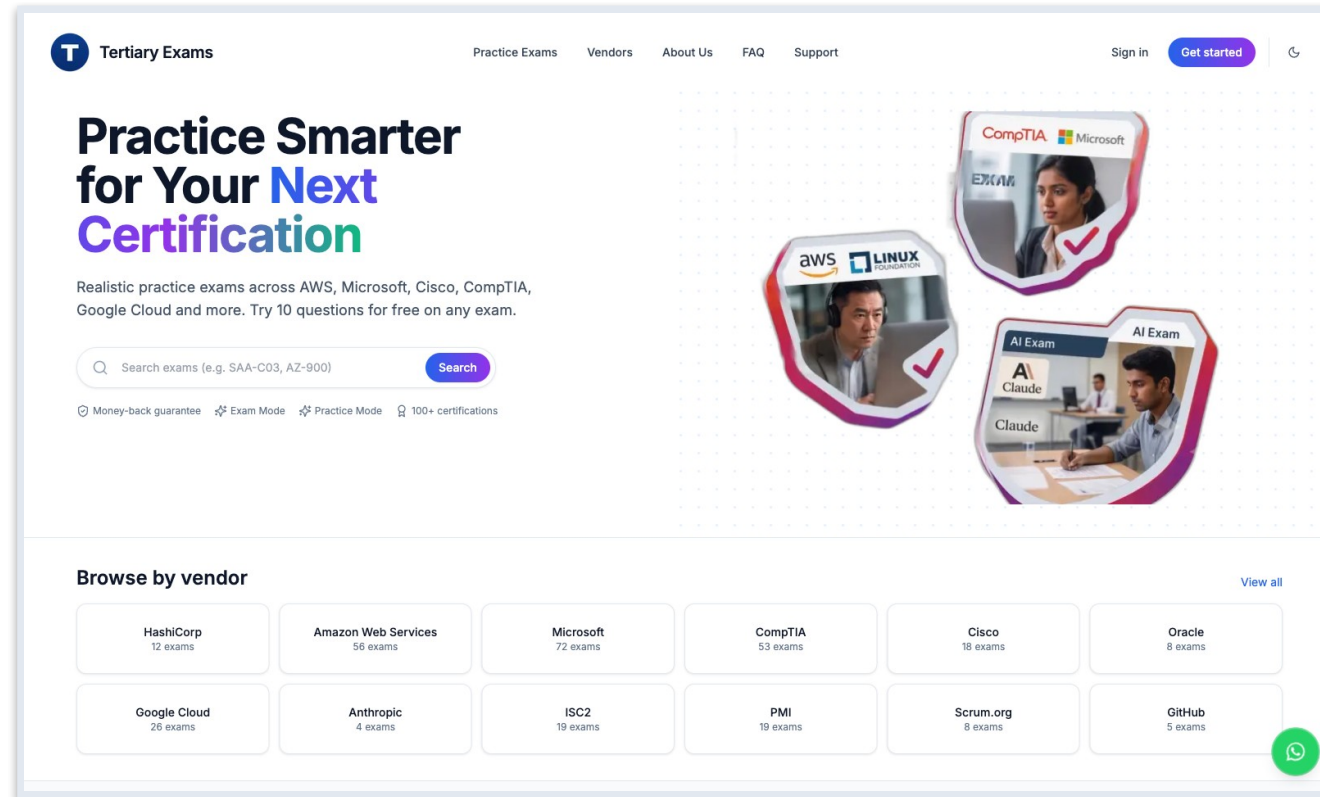
Assessment Flow



REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

Practice Before the Assessment



WHAT TO LOOK AT

Work through the practice questions at exams.tertiaryinfotech.com to rehearse the knowledge questions before you sit the Written Assessment.

Criteria for Funding

1

Attendance

A minimum attendance rate of 75%, based on the SSG Digital Attendance record.

2

Assessment

Complete both assessment components and be assessed as 'Competent'.

3

Digital attendance

Scan the SSG QR code for AM, PM and Assessment on every training day.

4

TRAQOM survey

Complete the mandatory TRAQOM course feedback survey on the LMS.

Set Up Before We Start

1

Microsoft 365 account

Sign in with the training account your trainer issues — it carries the Power Platform licence.

2

The course environment

Switch the environment picker to TGS-2022015539-Applications Integration with Power Apps and Power Automate.

3

Power Automate

Open make.powerautomate.com — this is where you build every flow.

4

Power Apps

Open make.powerapps.com — this is where you build every app.

5

Lab data

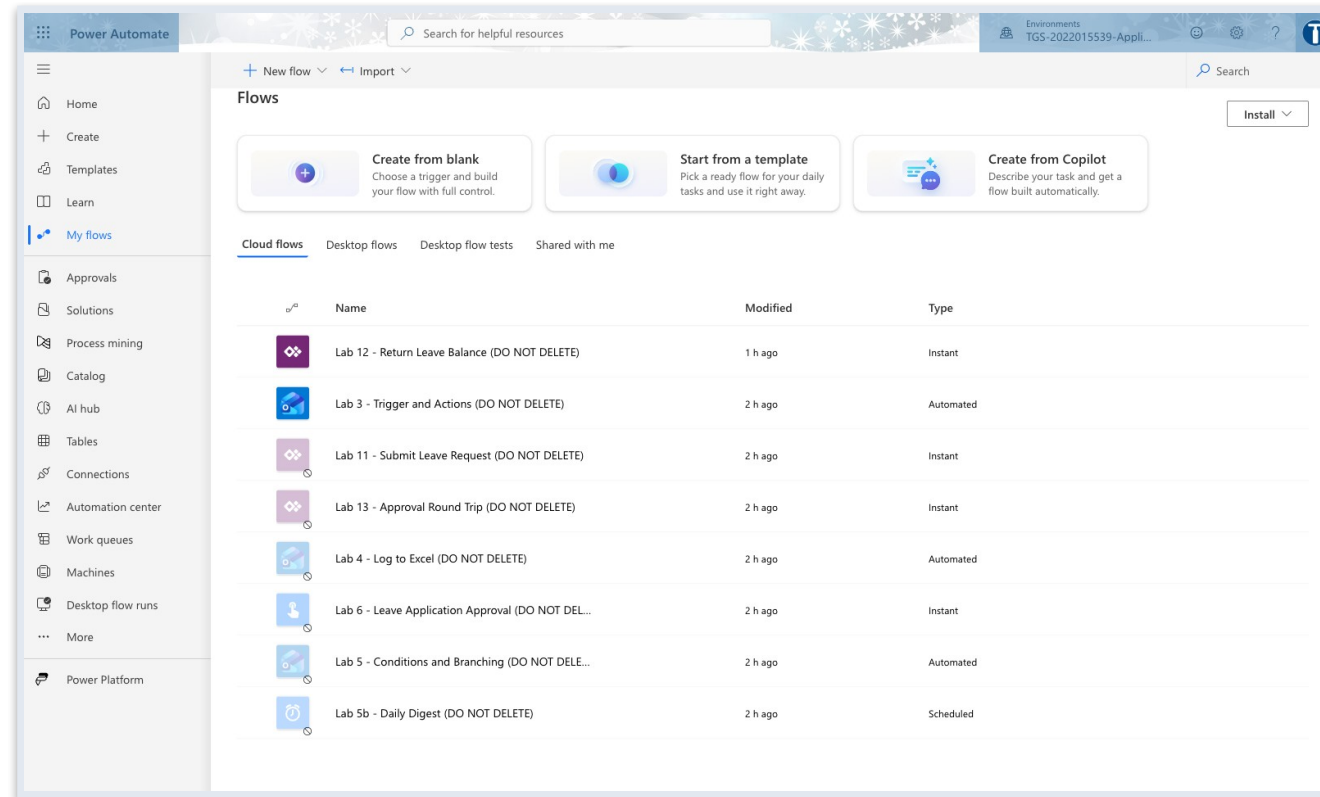
Download the lab workbooks from the LMS and upload them to your OneDrive for Business.

6

Microsoft Teams

Approvals arrive in Teams, so make sure you can sign in to it.

Your Lab Environment



WHAT TO LOOK AT

Every flow you build is named 'Lab N - ... (DO NOT DELETE)'. The environment picker at the top right must read the course environment before you build anything.



CORE CONCEPTS

Application Integration and the Power Platform

WHY THIS COURSE

Integration is what happens between the systems.

Most business software is bought, not built. The value you add is making the pieces talk to each other — and middleware is the layer where that happens.

What is Application Integration?

1

Connecting systems

Making two or more applications exchange data and trigger each other's functions without a human retyping anything.

2

Why it matters

Every manual re-keying step is slow, costs money and introduces errors that are expensive to find later.

3

Point-to-point vs hub

Direct connections are quick but multiply; a hub (middleware) keeps the number of connections manageable.

4

What you integrate

Devices, databases, software and applications — the four categories named in the TSC.

Types of Middleware and Their Features

Type	What it does	Typical example	Strength
Integration platform (iPaaS)	Cloud service that connects apps with prebuilt connectors	Power Automate, Zapier, MuleSoft	Fast to build, no servers to run
Message broker	Queues messages between producers and consumers	Azure Service Bus, RabbitMQ	Decouples systems; survives outages
API gateway	Fronts and secures a set of APIs	Azure API Management	Central auth, throttling, versioning
ETL / data pipeline	Moves and reshapes data in bulk on a schedule	Azure Data Factory, SSIS	High volume, scheduled loads
RPA	Drives a user interface when there is no API	Power Automate Desktop	Reaches legacy systems with no API

WHEN IT MATTERS Power Platform is an iPaaS with RPA attached — that is why it can reach both modern SaaS APIs and legacy desktop applications.

The Microsoft Power Platform Family

1

Power Apps

Build custom business apps quickly with low code — canvas apps and model-driven apps.

2

Power Automate

Automate workflows between applications and services — the middleware in this course.

3

Power BI

Analyse and visualise data from across those connected systems.

4

Copilot Studio

Build conversational agents over the same data and actions.

5

Dataverse

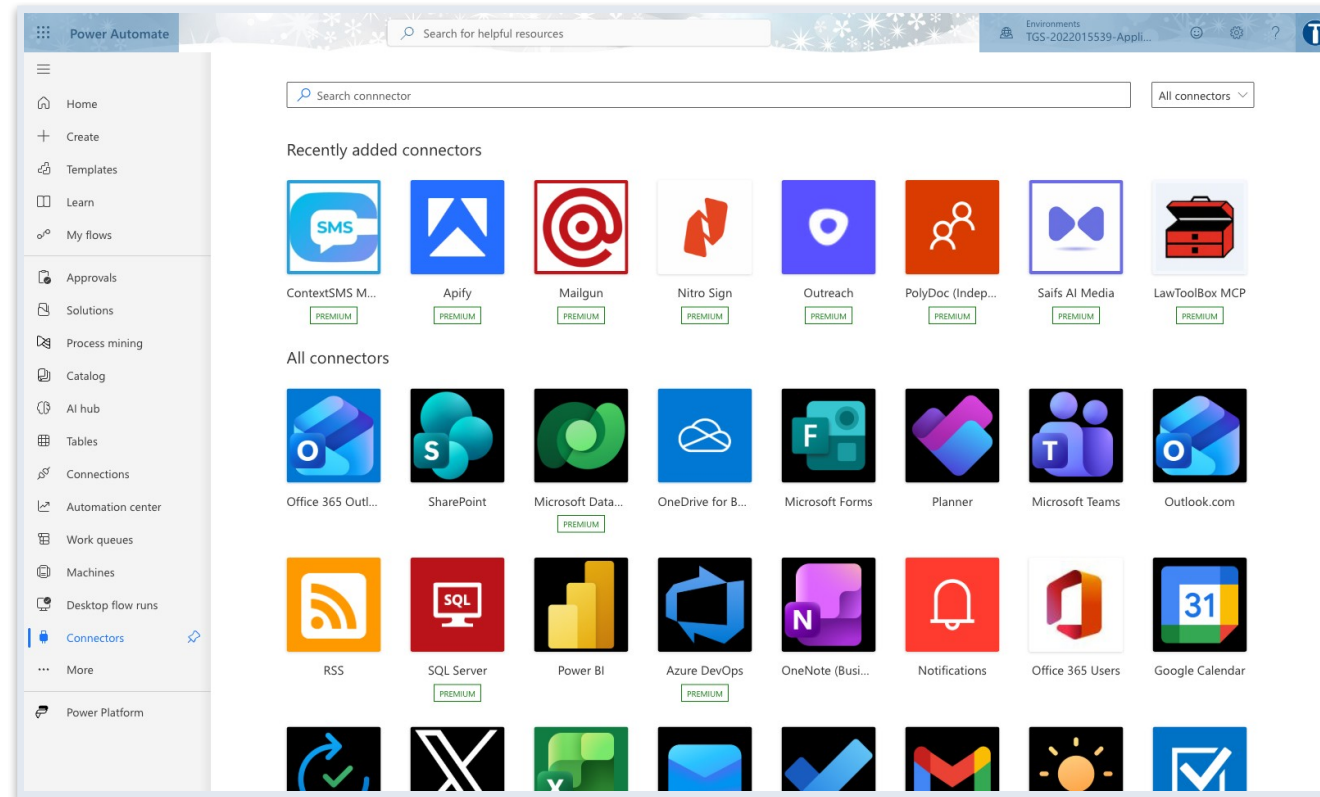
The secure, governed data store that underpins the whole platform.

6

Connectors

1,000+ prebuilt connectors plus custom connectors — the API surface that ties it all together.

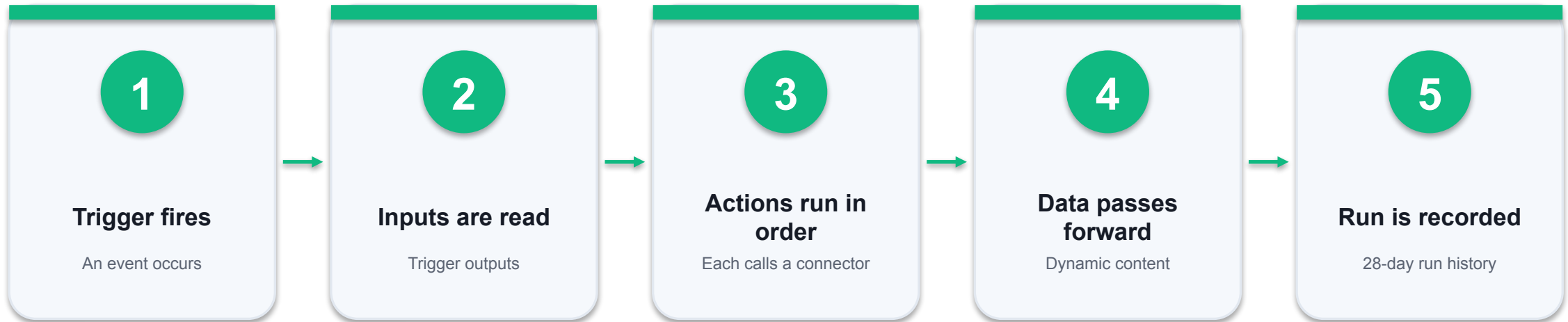
Connectors — the API Surface of the Platform



WHAT TO LOOK AT

A connector is a packaged API. Standard connectors are included; Premium connectors need a licence; a custom connector wraps any REST API you can reach.

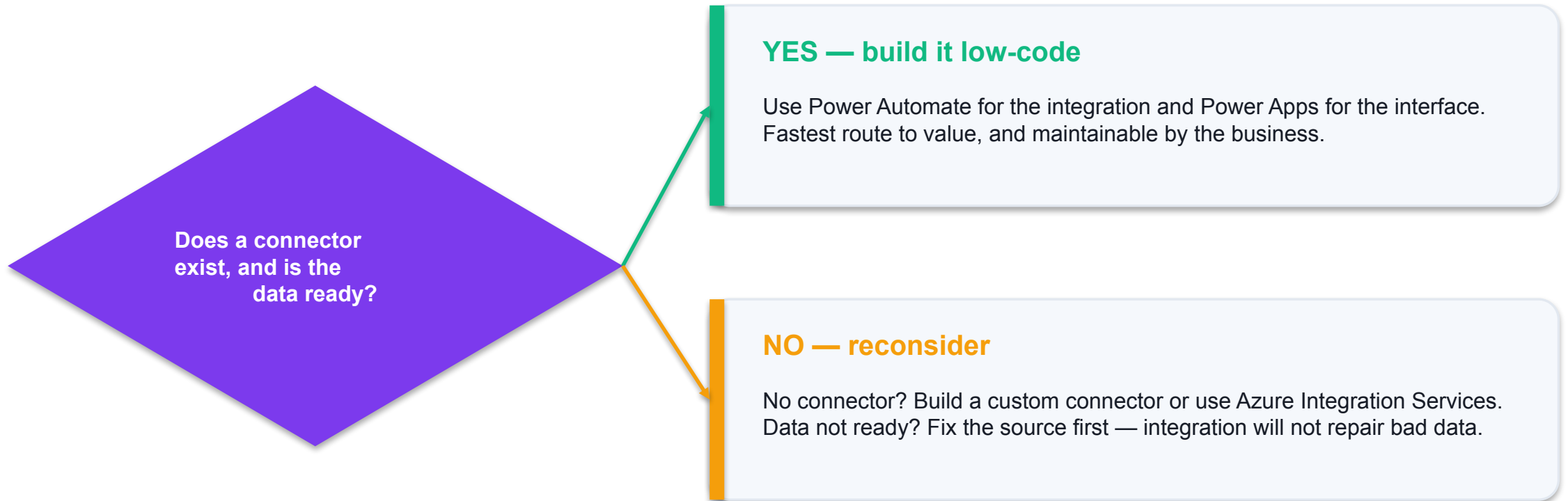
How a Cloud Flow Executes



THE POINT

A flow is a trigger plus an ordered set of actions. Everything you build in Topic 2 is a variation on this one shape.

Is Power Platform the Right Middleware Here?



A feasibility scan that approves everything is not a scan. Recording a defensible 'no' is as valuable as approving a build.

The Six Feasibility Criteria

1

Connector availability

Does a standard connector exist, is it Premium, or must you build a custom connector?

2

Data readiness

Is the source structured, does it have a stable key, is the Excel range a real Table?

3

Licensing

Premium connectors and Dataverse need the right plan — check before you design.

4

Security & governance

Who owns the data, which account runs the flow, does a DLP policy block the combination?

5

Performance

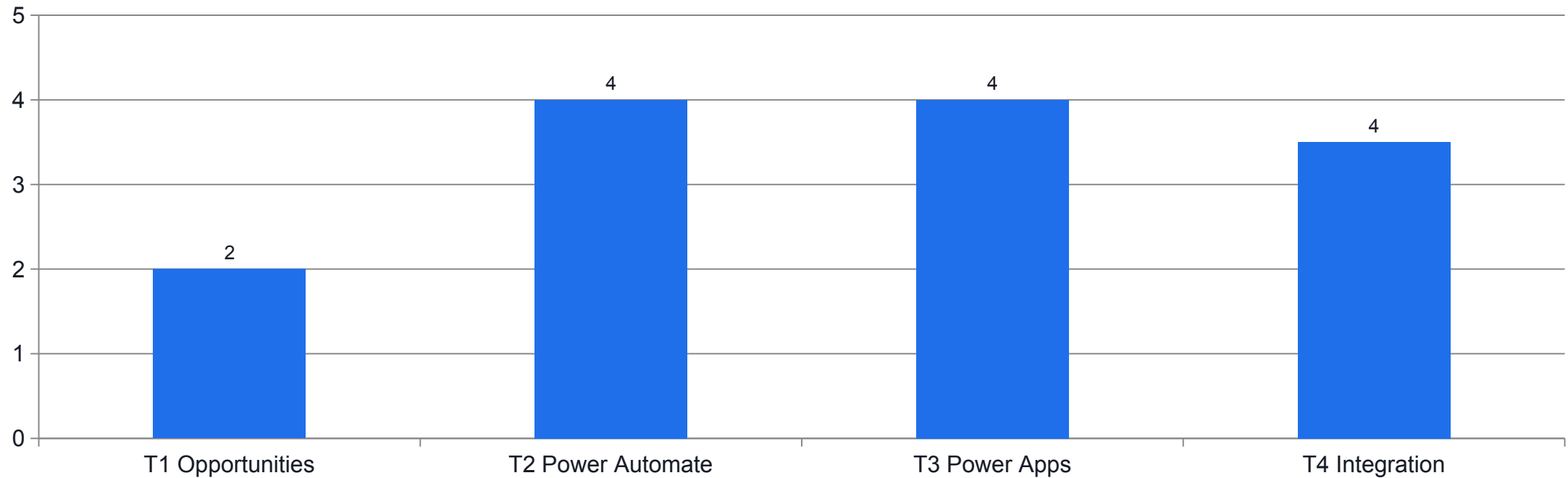
Row counts, delegation support and connector throttling limits.

6

Maintainability

Who supports this after you build it, and how will they know when it breaks?

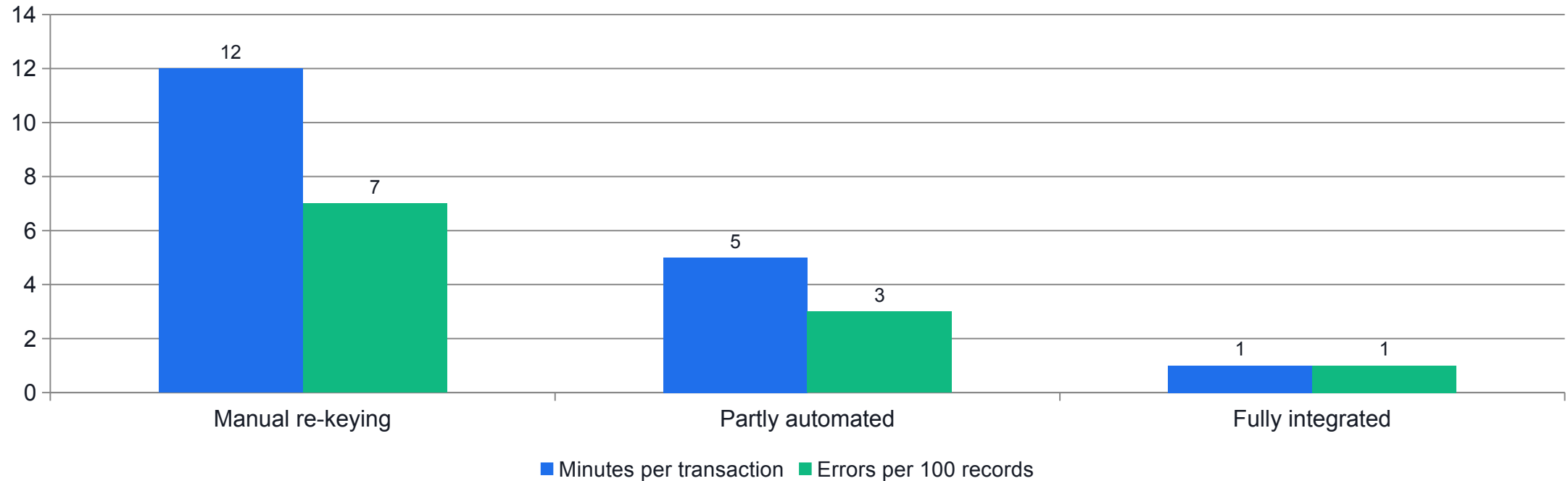
Where Course Time Goes — Topic Weighting



WHAT THE DATA SHOWS

Topics 2 and 3 carry the most time because that is where the practical ability assessed by the PP is built. Topic 4 is shorter but joins everything together — and is where the A7/A8 troubleshooting marks sit.

The Cost of Manual Re-Keying



WHAT THE DATA SHOWS

Indicative figures for teaching the trade-off, not a benchmark. The shape is what matters: integration cuts both handling time and error rate, and the error reduction is usually worth more than the time saved.

Canvas App vs Model-Driven App

Dimension	Canvas app	Model-driven app
Who designs the UI	You do — pixel by pixel	Generated from the data model
Data source	Any connector — Excel, SharePoint, SQL, APIs	Dataverse only
Best for	Task-focused apps with a specific look	Data-heavy, process-driven applications
Skill needed	Power Fx formulas, layout sense	Data modelling, forms and views
Build speed	Fast for small apps	Fast for large, standard CRUD apps

WHEN IT MATTERS This course builds canvas apps, because a canvas app can bind to any connector — which is what makes it an integration surface.

Cloud Flow Types

1

Automated

Starts when an event happens in a connected service — a mail arrives, a file is created, a row is added.

2

Instant

Started by a person, from a button, the mobile app, or from inside a canvas app.

3

Scheduled

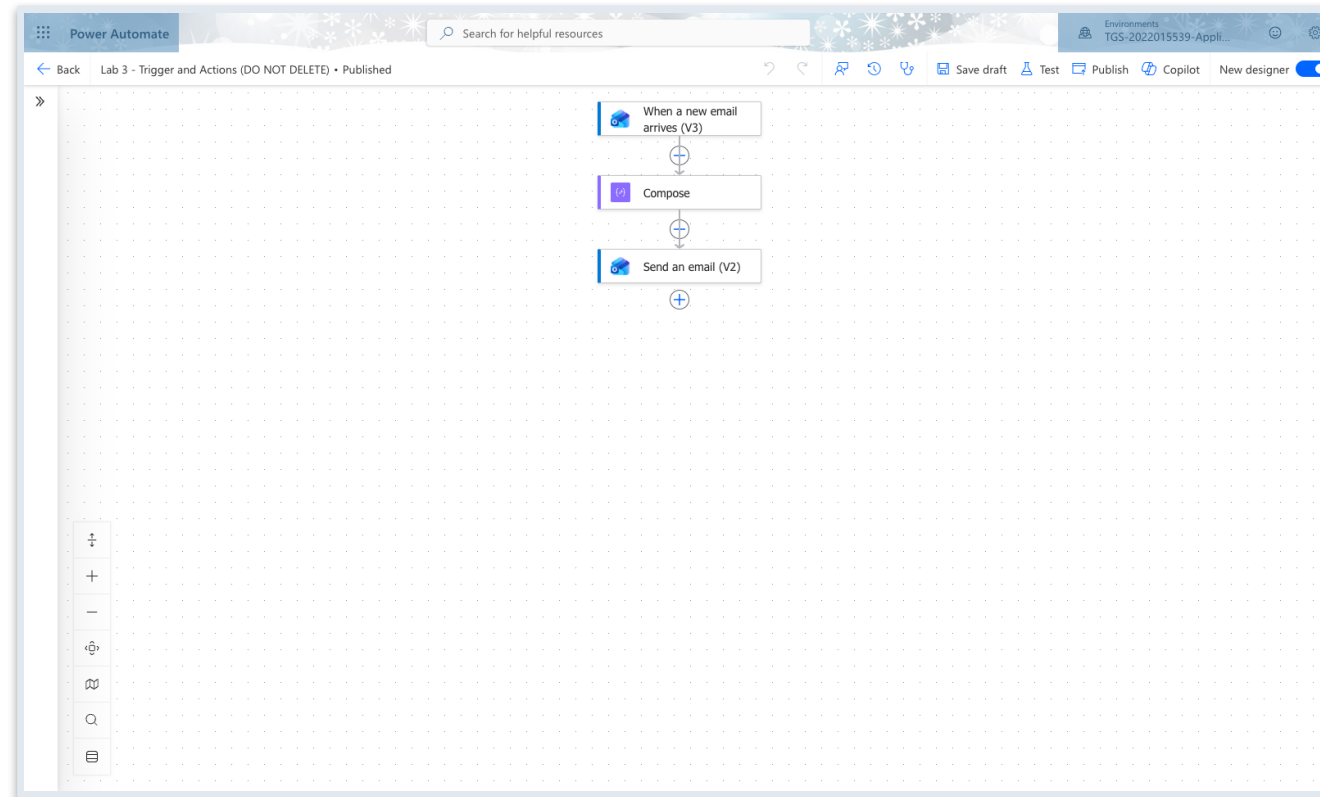
Runs on a recurrence — hourly, daily, weekly — for digests, syncs and housekeeping.

4

Desktop (RPA)

Drives a user interface for systems that have no API at all.

The Power Automate Designer



WHAT TO LOOK AT

The trigger sits at the top, actions run downward, and the + between them adds a step. The right-hand pane configures whichever action is selected.

Reading a Flow Definition

Every flow you build is this shape underneath — learn to read it once and the designer stops being a mystery.

```
"triggers": {
  "When_a_new_email_arrives_(V3)": {
    "type": "OpenApiConnectionNotification",
    "inputs": { "parameters": {
      "folderPath": "Inbox",
      "subjectFilter": "SERVICE" } } } },
"actions": {
  "Compose": { "type": "Compose",
    "inputs": "@{triggerOutputs()?['body/from']}" },
  "Send_an_email_(V2)": {
    "runAfter": { "Compose": ["Succeeded"] } } }
```

triggers

The single event that starts the run. A flow has exactly one.

subjectFilter

Narrows what fires the flow — cheaper and clearer than filtering later.

@{triggerOutputs()...}

Dynamic content: how data crosses from one action to the next.

runAfter

The dependency that orders the actions, and what to do when one fails.

Power Fx — the Formula Language

1

Excel-like

If you can write an Excel formula, you can write Power Fx. Same mental model, same functions.

2

Declarative

You set a property to a formula; it recalculates whenever its inputs change.

3

Binds controls to data

Items, Text, Visible, DisplayMode and OnSelect are the properties you will use most.

4

Delegation matters

Some functions run in the data source; others pull rows to the device. Know which.

The Three Classes of Integration Issue

Class	What it looks like	Typical cause	How you fix it
Technical	Flow fails with 401/403; connection shows an error	Expired or revoked connection, wrong account, DLP policy	Re-authenticate the connection; check DLP; use a service account
Compatibility	Schema validation error; wrong data in the target field	Type mismatch — text passed where a number is expected; date format	Coerce the type at the call site; agree a contract for the payload
Performance	Blue delegation warning; only the first rows appear; timeouts	Non-delegable function; too many rows; connector throttling	Rewrite to a delegable filter; reduce rows at source; batch the calls

WHEN IT MATTERS Lab 14 makes you provoke one of each on purpose — diagnosing a fault you caused is the fastest way to learn to diagnose one you did not.

K4 · THE CLASSIC PERFORMANCE TRAP

Delegation is the defect you will meet most.

When a formula cannot be delegated, Power Apps pulls only the first 500 rows to the device and filters there — so your app silently shows the wrong answer instead of failing loudly.

TOPIC 01

Opportunities for Using Power Platform Apps

Integration landscape · feasibility scan · Power Platform product family

Key Concepts — Opportunities for Using Power Platform Apps

1

Application integration connects disparate systems so data and functions flow between them without manual re-keying.

2

Middleware is the software layer that sits between applications and brokers that exchange — Power Platform is low-code integration middleware.

3

The Power Platform family: Power Apps (build), Power Automate (automate), Power BI (analyse), Copilot Studio (converse), Dataverse (store).

Key Concepts — Opportunities for Using Power Platform Apps (continued)

1

A feasibility scan weighs business value, data readiness, connector availability, licensing, security and governance before you build.

2

Connectors are the API surface of the platform: 1,000+ prebuilt connectors plus custom connectors for any REST API.

Hands-On Labs — Opportunities for Using Power Platform Apps

Lab 1

- Integration Opportunity Scan

Lab 2

- Feasibility Scan and Middleware Selection

Integration Opportunity Scan

LAB 1

K1 Types of middleware and their features; A1 Identify opportunities for creating connections among various devices, databases, software and applications.

Working from the KinetEco service-desk scenario, map the organisation's current manual process end to end and mark every point where data is re-keyed between systems. Each re-keying point is an integration opportunity. Classify each candidate by the systems it joins (device, database, software, application) and score it for volume, error rate and business impact.

YOU'LL BUILD

A completed Integration Opportunity Register listing at least six candidate integrations, each classified and scored.

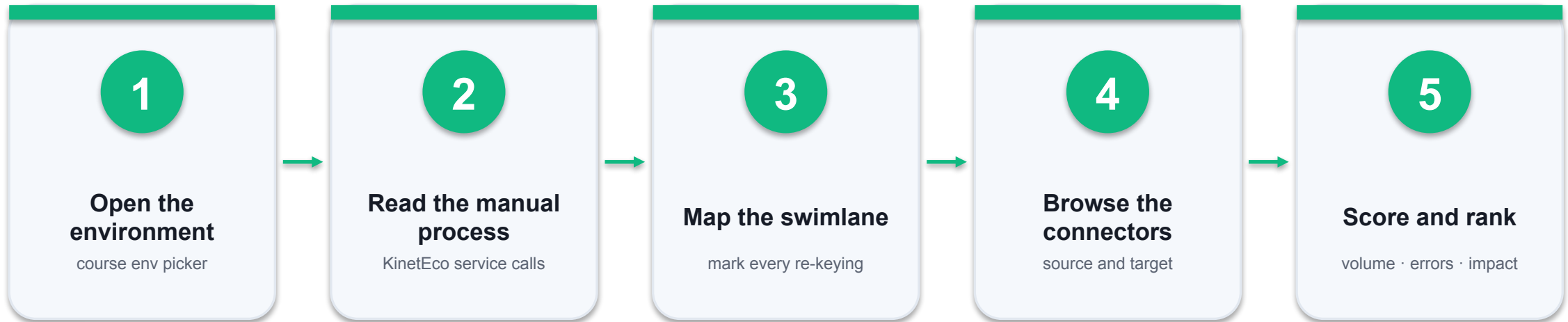
TOOLCHAIN

Power Platform admin center, Excel, Power Automate connector catalogue

DONE WHEN

Your register lists at least six candidates, each naming a source system, a target system, a connector and a score. The top-ranked candidate should be the service-call intake — the one you automate in Lab 3.

How Lab 1 Runs



THE POINT

K1 Types of middleware and their features; A1 Identify opportunities for creating connections among various devices, databases, software and applications.

Integration Opportunity Scan

✓ Expected result

Your register lists at least six candidates, each naming a source system, a target system, a connector and a score. The top-ranked candidate should be the service-call intake — the one you automate in Lab 3.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Feasibility Scan and Middleware Selection

LAB 2

A2 Perform feasibility scan and assessment to identify potential middleware to be used; K1 Types of middleware and their features.

Take the top three opportunities from Lab 1 and run a structured feasibility scan against six criteria: connector availability, data readiness, licensing, security and governance, performance/delegation, and maintainability. Decide for each whether Power Platform is the right middleware — and be prepared to justify a 'no'.

YOU'LL BUILD

A Feasibility Assessment Matrix scoring three candidates against six criteria, with a documented Build / Defer / Reject decision and justification for each.

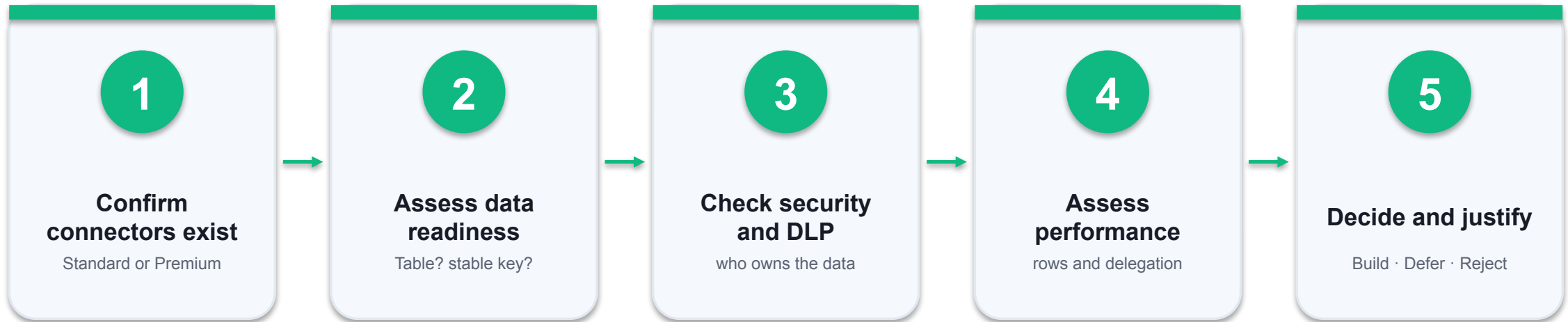
TOOLCHAIN

Power Platform admin center, connector catalogue, Microsoft licensing documentation

DONE WHEN

Each of the three candidates has six scores, a total, and a decision with justification. At least one candidate should be Deferred or Rejected with a defensible reason — a scan where everything passes is not a scan.

How Lab 2 Runs



THE POINT

A2 Perform feasibility scan and assessment to identify potential middleware to be used; K1 Types of middleware and their features.

Feasibility Scan and Middleware Selection

✓ Expected result

Each of the three candidates has six scores, a total, and a decision with justification. At least one candidate should be Deferred or Rejected with a defensible reason — a scan where everything passes is not a scan.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Recap — Opportunities for Using Power Platform Apps

- You can now: K1 Types of middleware and their features; A1 Identify opportunities for creating connections among various devices, databases, software and applications.
- You can now: A2 Perform feasibility scan and assessment to identify potential middleware to be used; K1 Types of middleware and their features.

TOPIC 02

Power Automate

Cloud flows · triggers and actions · conditions · approvals · testing

Key Concepts — Power Automate

1

A cloud flow is a trigger plus a sequence of actions; the trigger is the event that starts the flow.

2

Flow types: automated (event-driven), instant (manual/button), scheduled (recurrence), and desktop (RPA).

3

Dynamic content passes output from one action into the input of the next — this is how data crosses application boundaries.

Key Concepts — Power Automate (continued)

1

Conditions, loops and scopes give a flow its control logic; expressions add computation.

2

Testing is a first-class step: Flow checker, Test, and the 28-day run history are how you prove a connection works.

Hands-On Labs — Power Automate

Labs 3–4

- Trigger and Actions
- Log to Excel

Labs 5–6

- Conditions and Branching
- Leave Application Approval

Trigger and Actions

LAB 3

K2 Proper usage of middleware; A3 Utilise middleware to integrate data and functions across application programs.

Build your first cloud flow: an automated flow that fires when a new service-call item arrives and sends a formatted notification. This is the atom of every integration — one trigger, one or more actions, and dynamic content carrying data between them.

YOU'LL BUILD

Flow 'Lab 3 - Trigger and Actions (DO NOT DELETE)' — an automated cloud flow with a trigger, a compose step and a notification action, tested with a successful run.

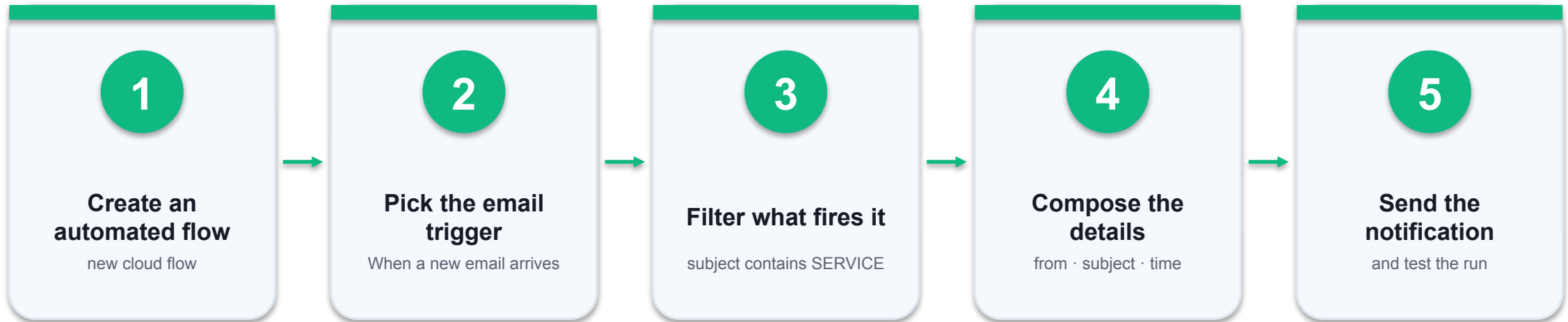
TOOLCHAIN

Power Automate, Office 365 Outlook, Compose

DONE WHEN

The run history shows one Succeeded run. Expanding the Compose step shows the sender, subject and timestamp carried from the trigger, and the notification email arrives in your inbox.

How Lab 3 Runs



THE POINT

K2 Proper usage of middleware; A3 Utilise middleware to integrate data and functions across application programs.

Trigger and Actions

✓ Expected result

The run history shows one Succeeded run. Expanding the Compose step shows the sender, subject and timestamp carried from the trigger, and the notification email arrives in your inbox.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Log to Excel

LAB 4

A3 Utilise middleware to integrate data and functions across application programs; A5 Perform tests and checks on the connections between disparate application programs.

Extend Lab 3 so the service call is not merely announced but recorded. The flow writes a row into an Excel Online table in OneDrive, turning a notification into a durable integration between a mail system and a data store.

YOU'LL BUILD

Flow 'Lab 4 - Log to Excel (DO NOT DELETE)' writing one row per service call into a formatted Excel Table, verified by inspecting the workbook.

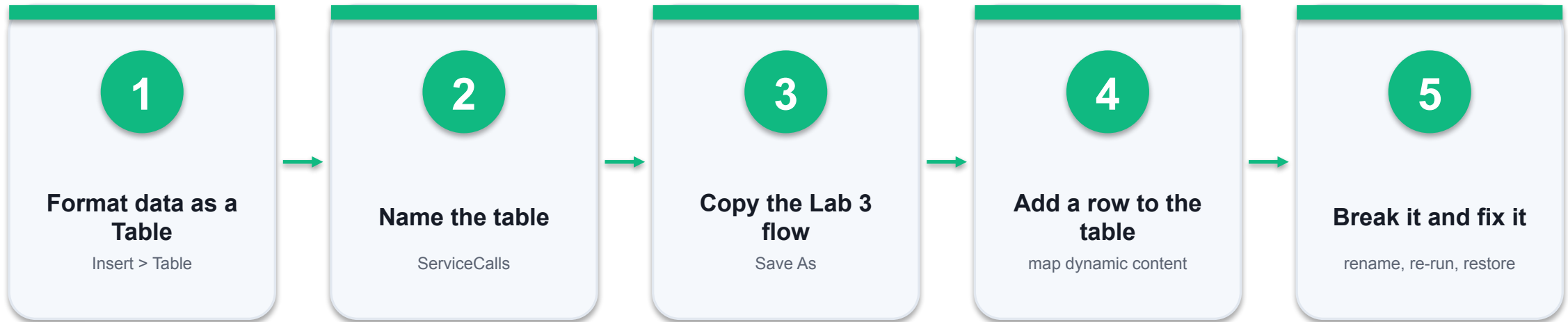
TOOLCHAIN

Power Automate, Office 365 Outlook, Excel Online (Business), OneDrive for Business

DONE WHEN

A new row appears in the ServiceCalls table for each test email. The deliberate rename produces a clear failure in run history, and restoring the name returns the flow to Succeeded.

How Lab 4 Runs



THE POINT

A3 Utilise middleware to integrate data and functions across application programs; A5 Perform tests and checks on the connections between disparate application programs.

Log to Excel

✓ Expected result

A new row appears in the ServiceCalls table for each test email. The deliberate rename produces a clear failure in run history, and restoring the name returns the flow to Succeeded.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Conditions and Branching

LAB 5

K2 Proper usage of middleware; A3 Utilise middleware to integrate data and functions across application programs.

Real integrations rarely run in a straight line. Add a Condition so urgent service calls are escalated while routine ones are only logged, then add a scheduled digest flow that summarises the day's calls.

YOU'LL BUILD

Flow 'Lab 5 - Conditions and Branching (DO NOT DELETE)' with a working If yes / If no branch, plus a scheduled recurrence flow producing a daily digest.

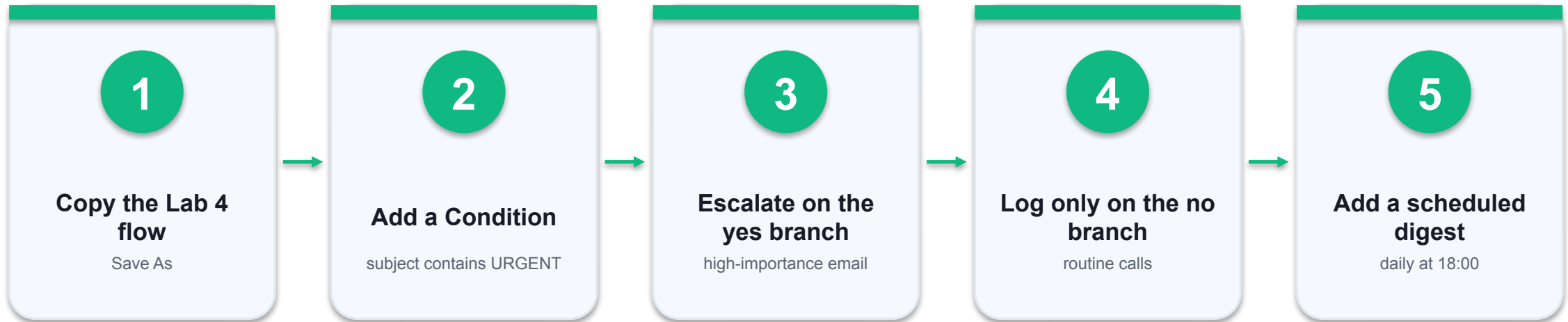
TOOLCHAIN

Power Automate, Office 365 Outlook, Excel Online (Business), Recurrence

DONE WHEN

Run history shows the URGENT test taking the If yes branch and the routine test taking If no. The digest flow sends one email containing an HTML table of the logged calls.

How Lab 5 Runs



THE POINT

K2 Proper usage of middleware; A3 Utilise middleware to integrate data and functions across application programs.

Conditions and Branching

✓ Expected result

Run history shows the URGENT test taking the If yes branch and the routine test taking If no. The digest flow sends one email containing an HTML table of the logged calls.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Leave Application Approval

LAB 6

A3 Utilise middleware to integrate data and functions across application programs; A5 Perform tests and checks on the connections between disparate application programs.

Add the human to the loop. Build an approval flow where a leave request is routed to an approver, the decision branches the flow, and the outcome is written back and notified — the canonical human-in-the-loop integration pattern.

YOU'LL BUILD

Flow 'Lab 6 - Leave Application Approval (DO NOT DELETE)' with a Start and wait for an approval action, an outcome condition, and write-back to Excel.

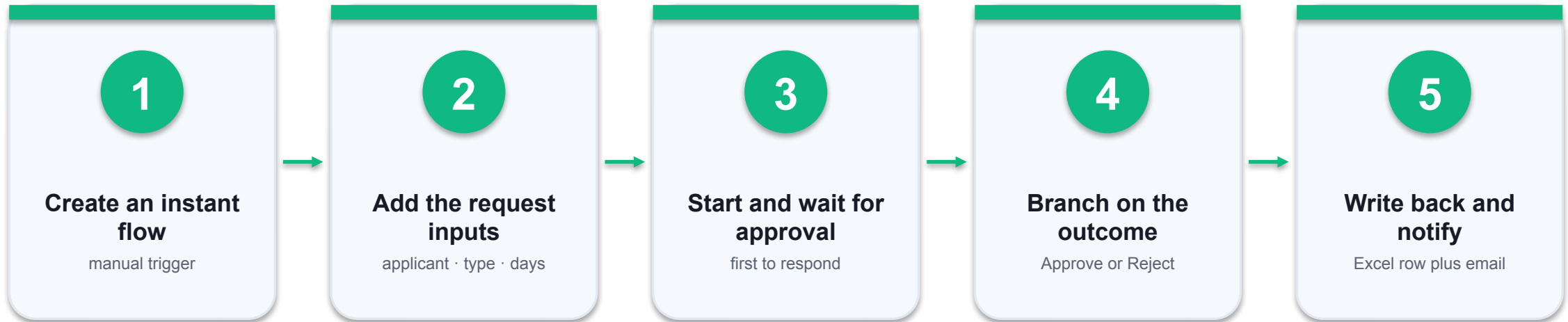
TOOLCHAIN

Power Automate, Approvals, Office 365 Outlook, Excel Online (Business), Microsoft Teams

DONE WHEN

Two runs complete: one Approved and one Rejected. Each produced the correct branch, the correct email, and a correctly-statused row in the LeaveLog table.

How Lab 6 Runs



THE POINT

A3 Utilise middleware to integrate data and functions across application programs; A5 Perform tests and checks on the connections between disparate application programs.

Leave Application Approval

✓ Expected result

Two runs complete: one Approved and one Rejected. Each produced the correct branch, the correct email, and a correctly-statused row in the LeaveLog table.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Recap — Power Automate

- You can now: K2 Proper usage of middleware; A3 Utilise middleware to integrate data and functions across application programs.
- You can now: A3 Utilise middleware to integrate data and functions across application programs; A5 Perform tests and checks on the connections between disparate application programs.

TOPIC 03

Power Apps

Canvas apps · data sources · controls and formulas · publishing and sharing

Key Concepts — Power Apps

1

Canvas apps give pixel-level control of the UI; model-driven apps generate the UI from the data model.

2

Apps run across platforms — browser, iOS, Android, Teams and embedded in SharePoint or Power BI.

3

A data source is reached through a connector; the same app can bind to Dataverse, Excel, SharePoint or a REST API.

Key Concepts — Power Apps (continued)

1

Power Fx is the Excel-like formula language that binds controls to data and behaviour.

2

Publish, version and share are the app lifecycle: every save creates a version you can restore.

Hands-On Labs — Power Apps

Labs 7–8

- Canvas App from Excel Data
- Blank Canvas App with Power Fx

Labs 9–10

- Custom Connector and API Integration
- Cross-Platform Verification and Sharing

Canvas App from Excel Data

LAB 7

K3 Different types of platforms on which applications run; A4 Support API-level integration.

Create your first canvas app over the Excel table you built in Lab 4. Power Apps generates a three-screen browse/detail/edit app; you then read how each screen is wired so the generated app becomes something you can modify rather than a black box.

YOU'LL BUILD

Canvas app 'Lab 7 - Canvas App from Excel (DO NOT DELETE)' — a working three-screen app bound to the ServiceCalls Excel table, saved and published.

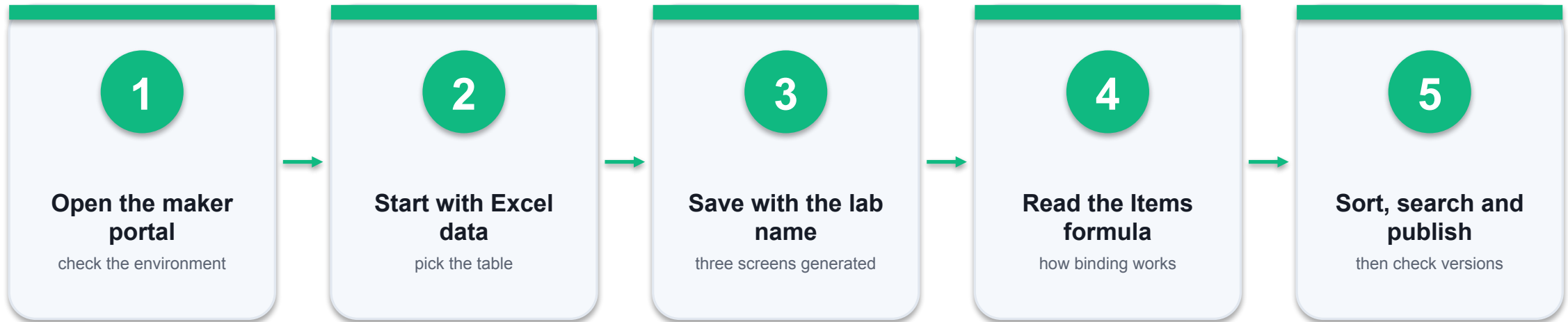
TOOLCHAIN

Power Apps Studio, Excel Online (Business), OneDrive for Business

DONE WHEN

The app browses, filters and sorts real service-call data; adding a record through the app writes a new row to the Excel table, and the app appears in the Apps list as published.

How Lab 7 Runs



THE POINT

K3 Different types of platforms on which applications run; A4 Support API-level integration.

Canvas App from Excel Data

✓ Expected result

The app browses, filters and sorts real service-call data; adding a record through the app writes a new row to the Excel table, and the app appears in the Apps list as published.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Blank Canvas App with Power Fx

LAB 8

K3 Different types of platforms on which applications run; A6 Verify proper functioning of modules and applications across multiple or integrated platforms.

A generated app hides the mechanics. Build one from an empty screen so every control, variable and formula is yours: a leave-request form with validation, variables and navigation, which becomes the front end you wire to a flow in Topic 4.

YOU'LL BUILD

Canvas app 'Lab 8 - Leave Request App (DO NOT DELETE)' — a two-screen app built from blank with validated inputs, variables and navigation.

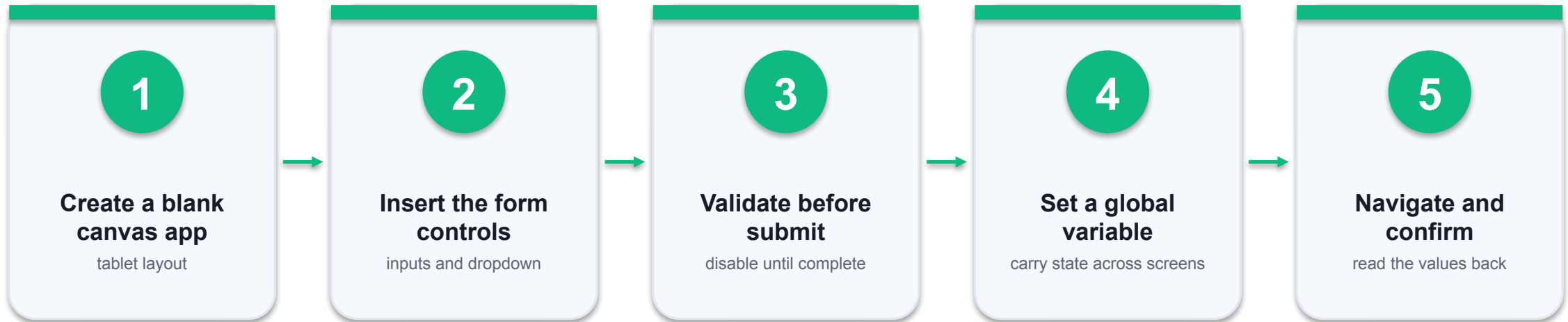
TOOLCHAIN

Power Apps Studio, Power Fx

DONE WHEN

The submit button stays disabled until the required fields are filled; submitting navigates to the confirmation screen showing the entered values; App checker reports no errors.

How Lab 8 Runs



THE POINT

K3 Different types of platforms on which applications run; A6 Verify proper functioning of modules and applications across multiple or integrated platforms.

Blank Canvas App with Power Fx

✓ Expected result

The submit button stays disabled until the required fields are filled; submitting navigates to the confirmation screen showing the entered values; App checker reports no errors.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Custom Connector and API Integration

LAB 9

K5 Functions of Application Programming Interfaces (APIs); A4 Support API-level integration.

Connectors are the platform's API layer. Where no prebuilt connector exists you build a custom one. Create a custom connector over a public REST API, define its action and response schema, test it, and consume it from a canvas app.

YOU'LL BUILD

Custom connector 'Lab 9 - Public Holiday API (DO NOT DELETE)' plus canvas app 'Lab 9 - API Consumer App (DO NOT DELETE)' displaying live API data.

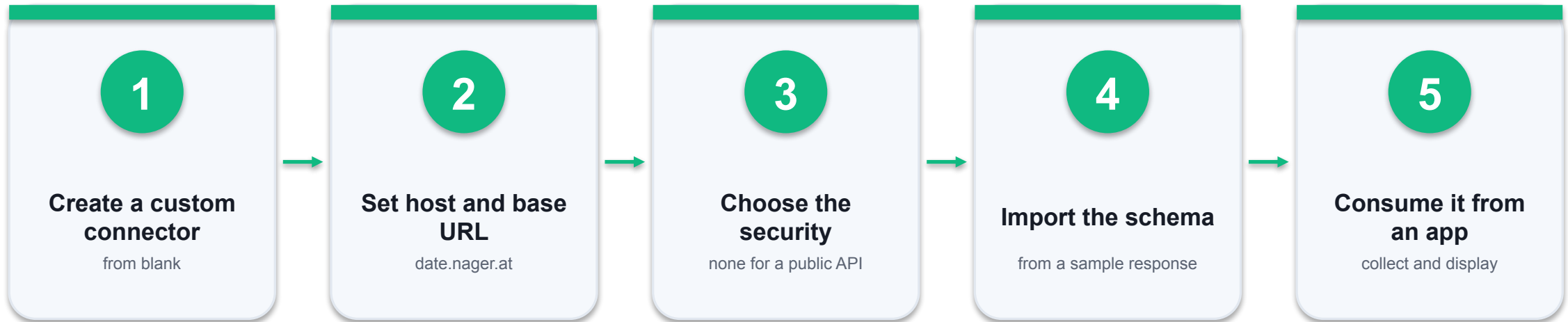
TOOLCHAIN

Power Apps custom connectors, REST, JSON, Power Fx

DONE WHEN

The connector Test tab returns HTTP 200 with a JSON array of holidays. In the app, pressing the button fills the gallery with live public-holiday data from the external API.

How Lab 9 Runs



THE POINT

K5 Functions of Application Programming Interfaces (APIs); A4 Support API-level integration.

Custom Connector and API Integration

✓ Expected result

The connector Test tab returns HTTP 200 with a JSON array of holidays. In the app, pressing the button fills the gallery with live public-holiday data from the external API.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Cross-Platform Verification and Sharing

LAB 10

A6 Verify proper functioning of modules and applications across multiple or integrated platforms; K3 Different types of platforms on which applications run.

An app that only works on the maker's screen is not integrated. Verify your apps run correctly in the browser, on the mobile player and embedded in Teams, then share them with the right permissions and record the results in a test matrix.

YOU'LL BUILD

A completed Cross-Platform Verification Matrix for Labs 7-9, plus the apps shared and one app embedded in Microsoft Teams.

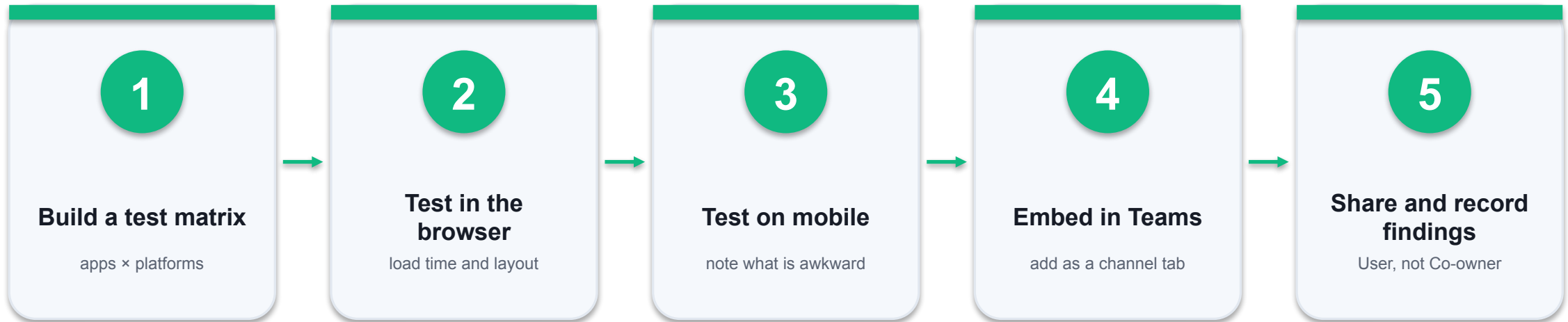
TOOLCHAIN

Power Apps mobile, Microsoft Teams, Power Apps sharing

DONE WHEN

Every cell of the matrix is filled for all three platforms, with at least three real defects or limitations recorded, and a classmate can run your shared app.

How Lab 10 Runs



THE POINT

A6 Verify proper functioning of modules and applications across multiple or integrated platforms; K3 Different types of platforms on which applications run.

Cross-Platform Verification and Sharing

✓ Expected result

Every cell of the matrix is filled for all three platforms, with at least three real defects or limitations recorded, and a classmate can run your shared app.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Recap — Power Apps

- You can now: K3 Different types of platforms on which applications run; A4 Support API-level integration.
- You can now: K3 Different types of platforms on which applications run; A6 Verify proper functioning of modules and applications across multiple or integrated platforms.
- You can now: K5 Functions of Application Programming Interfaces (APIs); A4 Support API-level integration.
- You can now: A6 Verify proper functioning of modules and applications across multiple or integrated platforms; K3 Different types of platforms on which applications run.

TOPIC 04

Integrate Power Apps and Power Automate

Calling flows from apps · approvals · returning data · troubleshooting integration

Key Concepts — Integrate Power Apps and Power Automate

1

A canvas app calls a flow with the Power Apps (V2) trigger; parameters are typed and passed from controls.

2

Respond to a Power App or flow returns data back to the app, closing the round trip.

3

Approvals combine a flow, an approver and a decision branch — the classic human-in-the-loop integration.

Key Concepts — Integrate Power Apps and Power Automate (continued)

1

Integration issues cluster into technical (auth, connector), compatibility (data type, format) and performance (delegation, throughput).

2

Delegation is the single most common Power Apps performance defect — know the warning and the mitigations.

Hands-On Labs — Integrate Power Apps and Power Automate

Labs 11–12

- Call a Flow from a Canvas App
- Return Data from a Flow to the App

Labs 13–14

- End-to-End Approval App
- Troubleshoot and Optimise the Integration

Call a Flow from a Canvas App

LAB 11

A4 Support API-level integration; A3 Utilise middleware to integrate data and functions across application programs.

Wire the Lab 8 app to a flow. The button no longer just sets a variable — it invokes a Power Automate flow with typed parameters, so the app becomes a front end to an integration rather than a form in isolation.

YOU'LL BUILD

Flow 'Lab 11 - Submit Leave Request (DO NOT DELETE)' triggered from the app, and the Lab 8 app updated to call it with typed parameters.

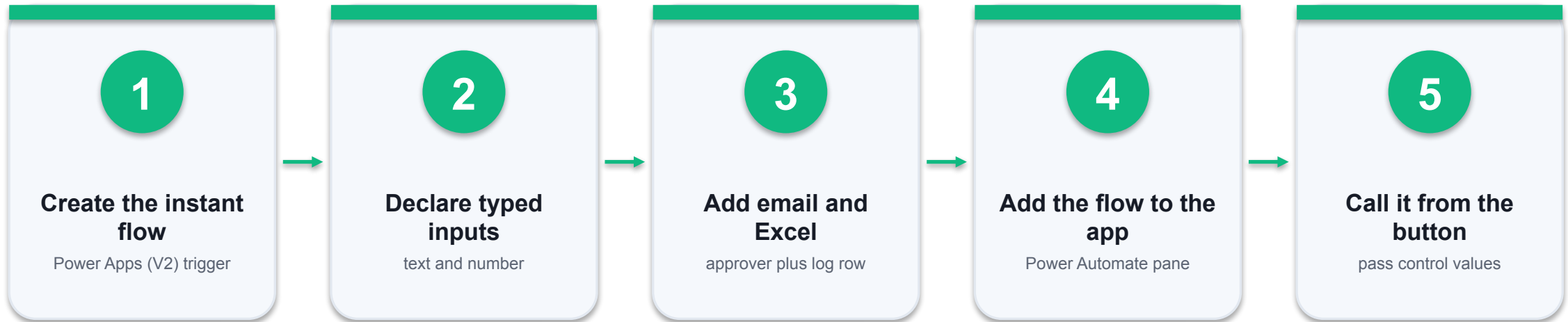
TOOLCHAIN

Power Apps, Power Automate, Power Apps (V2) trigger, Office 365 Outlook

DONE WHEN

Submitting from the app produces a Succeeded flow run whose trigger inputs exactly match what you typed, plus an approver email and a new LeaveLog row.

How Lab 11 Runs



THE POINT

A4 Support API-level integration; A3 Utilise middleware to integrate data and functions across application programs.

Call a Flow from a Canvas App

✓ Expected result

Submitting from the app produces a Succeeded flow run whose trigger inputs exactly match what you typed, plus an approver email and a new LeaveLog row.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Return Data from a Flow to the App

LAB 12

A4 Support API-level integration; A6 Verify proper functioning of modules and applications across multiple or integrated platforms.

So far data has flowed one way. Close the round trip with 'Respond to a Power App or flow' so the flow returns a result the app displays — the pattern that lets an app use server-side logic, secured credentials or external APIs it cannot reach directly.

YOU'LL BUILD

Flow 'Lab 12 - Return Leave Balance (DO NOT DELETE)' returning typed values, and the app displaying the returned balance and reference number.

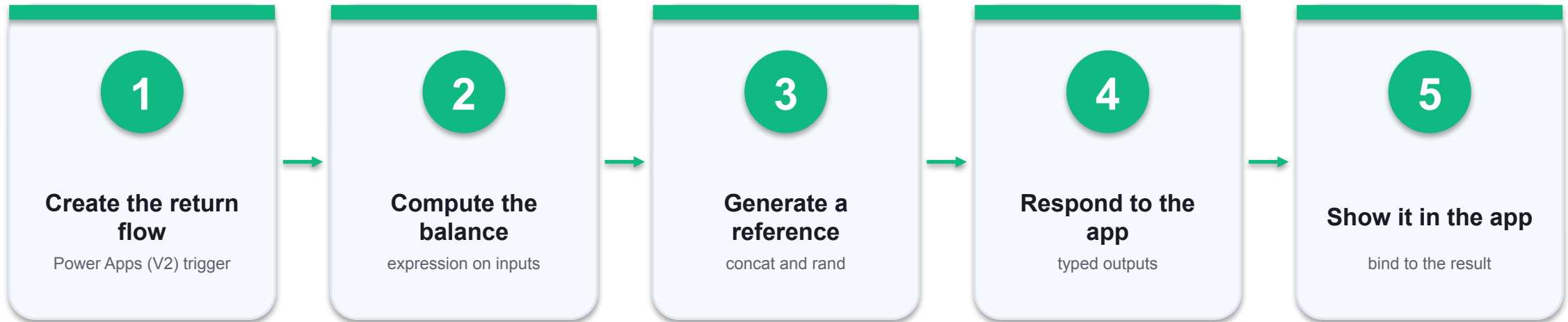
TOOLCHAIN

Power Apps, Power Automate, Respond to a Power App or flow, Excel Online (Business)

DONE WHEN

The confirmation screen displays a reference number and remaining balance produced by the flow, and those values match the 'Respond to a Power App or flow' outputs in run history.

How Lab 12 Runs



THE POINT

A4 Support API-level integration; A6 Verify proper functioning of modules and applications across multiple or integrated platforms.

Return Data from a Flow to the App

✓ Expected result

The confirmation screen displays a reference number and remaining balance produced by the flow, and those values match the 'Respond to a Power App or flow' outputs in run history.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

End-to-End Approval App

LAB 13

A3 Utilise middleware to integrate data and functions across application programs; A5 Perform tests and checks on the connections between disparate application programs.

Assemble everything into one working solution: the app submits, a flow requests approval, the approver decides in Teams, the outcome is written back, and the app shows live status. This is the integrated system the PP assessment asks you to build.

YOU'LL BUILD

Flow 'Lab 13 - Approval Round Trip (DO NOT DELETE)' plus app 'Lab 13 - Leave Portal App (DO NOT DELETE)' showing live request status from the data source.

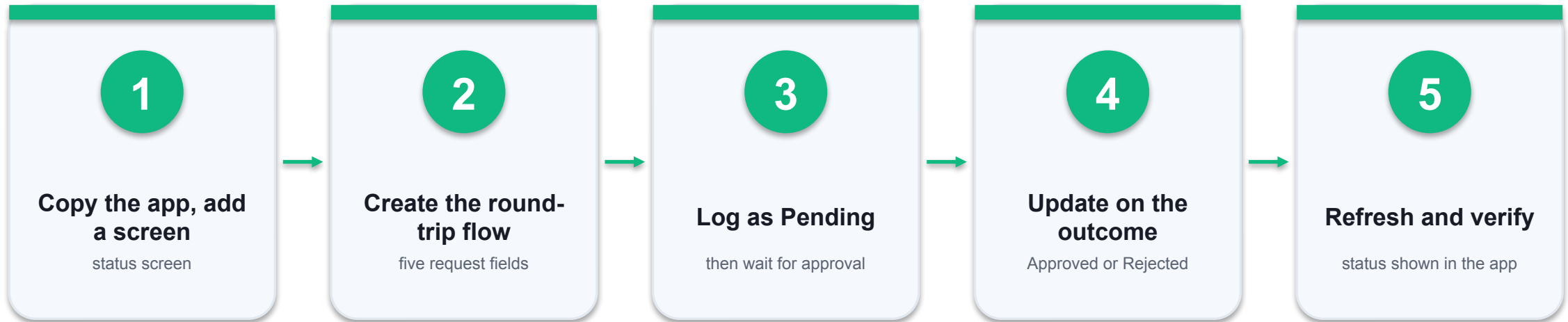
TOOLCHAIN

Power Apps, Power Automate, Approvals, Microsoft Teams, Excel Online (Business)

DONE WHEN

A request submitted in the app appears as Pending, moves to Approved or Rejected after the approver decides in Teams, and the app's status screen shows the change after Refresh.

How Lab 13 Runs



THE POINT

A3 Utilise middleware to integrate data and functions across application programs; A5 Perform tests and checks on the connections between disparate application programs.

End-to-End Approval App

✓ Expected result

A request submitted in the app appears as Pending, moves to Approved or Rejected after the approver decides in Teams, and the app's status screen shows the change after Refresh.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Troubleshoot and Optimise the Integration

LAB 14

K4 Potential technical, compatibility or performance issues in application integration; A7 Highlight technical, compatibility or performance issues; A8 Implement modifications to mitigate the issues identified.

Deliberately break your working solution in three ways — technical, compatibility and performance — then diagnose each from the evidence, fix it, and record the finding. This is the A7/A8 discipline the assessment tests, and it is far easier to learn on a system you built.

YOU'LL BUILD

A completed Integration Issue Log documenting three provoked defects with evidence, root cause and the implemented fix, plus the repaired solution.

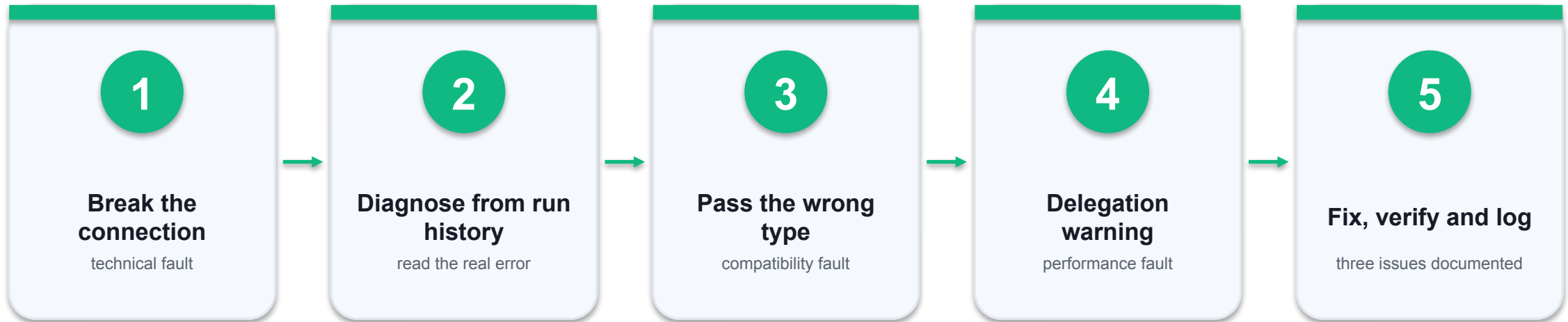
TOOLCHAIN

Power Automate run history, Power Apps App checker, Monitor, delegation warnings

DONE WHEN

The Issue Log documents three issues, one of each class, each with real evidence, a root cause and a verified fix. The end-to-end approval scenario runs successfully after all repairs.

How Lab 14 Runs



THE POINT

K4 Potential technical, compatibility or performance issues in application integration; A7 Highlight technical, compatibility or performance issues; A8 Implement modifications to mitigate the issues identified.

Troubleshoot and Optimise the Integration

✓ Expected result

The Issue Log documents three issues, one of each class, each with real evidence, a root cause and a verified fix. The end-to-end approval scenario runs successfully after all repairs.

IF IT DOESN'T WORK

Nothing happens

Check the .env file is in the labs folder and the key has no quotes or spaces.

Auth or 401 error

Re-copy the API key from AI Studio; confirm `GOOGLE_GENAI_USE_VERTEXAI=0`.

ModuleNotFoundError

Run `uv sync` again, and prefix commands with `uv run` so the venv is used.

Recap — Integrate Power Apps and Power Automate

- You can now: A4 Support API-level integration; A3 Utilise middleware to integrate data and functions across application programs.
- You can now: A4 Support API-level integration; A6 Verify proper functioning of modules and applications across multiple or integrated platforms.
- You can now: A3 Utilise middleware to integrate data and functions across application programs; A5 Perform tests and checks on the connections between disparate application programs.
- You can now: K4 Potential technical, compatibility or performance issues in application integration; A7 Highlight technical, compatibility or performance issues; A8 Implement modifications to mitigate the issues identified.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

LO1 · Identified opportunities

Mapped a manual process, built an Integration Opportunity Register and ran a six-criteria feasibility scan with defensible Build / Defer / Reject decisions.

2

LO2 · Utilised Power Automate

Built and tested automated, instant and scheduled cloud flows with triggers, actions, dynamic content, conditions, branching and approvals.

3

LO3 · Created Power Apps

Built canvas apps from data and from blank, wrote Power Fx, reached an external API through a custom connector, and verified the apps across browser, mobile and Teams.

4

LO4 · Improved apps with flows

Called flows from an app, returned data to close the round trip, built an end-to-end approval, then provoked, diagnosed and fixed technical, compatibility and performance defects.

Continue Your Learning

1

Microsoft Learn

learn.microsoft.com/power-platform — the official reference for every connector and function.

2

Power Fx reference

learn.microsoft.com/power-platform/power-fx — the formula language in full.

3

Rebuild the labs

Redo each lab from an empty flow until the trigger-action-test rhythm is automatic.

4

Extend the capstone

Add a second approver, a scheduled reminder, or a Power BI report over your LeaveLog.

5

Govern it

Learn environments, DLP policies and solutions before you deploy anything to production.

6

Community

The Power Platform community forums are the fastest route past a connector-specific problem.

Recommended Courses

1

WSQ - Business Process Automation with Power Automate and Copilot Studio Agents

2

WSQ - Data Analytics and Visualisation with Power BI

3

WSQ - Microsoft SharePoint for Collaboration

4

WSQ - Data Visualisation with Tableau

5

WSQ - Generative AI for Business Productivity

Support

- If you have any enquiries during or after the class, you can contact us below.
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg
- LMS / TMS: <https://lms-tms.tertiaryinfotech.com>

Assessment

1

Written Assessment (SAQ)

1 hour · open book · five short-answer knowledge questions covering K1–K5.

2

Practical Performance (PP)

1.5 hours · open book · four hands-on tasks covering A1–A8.

3

Digital attendance

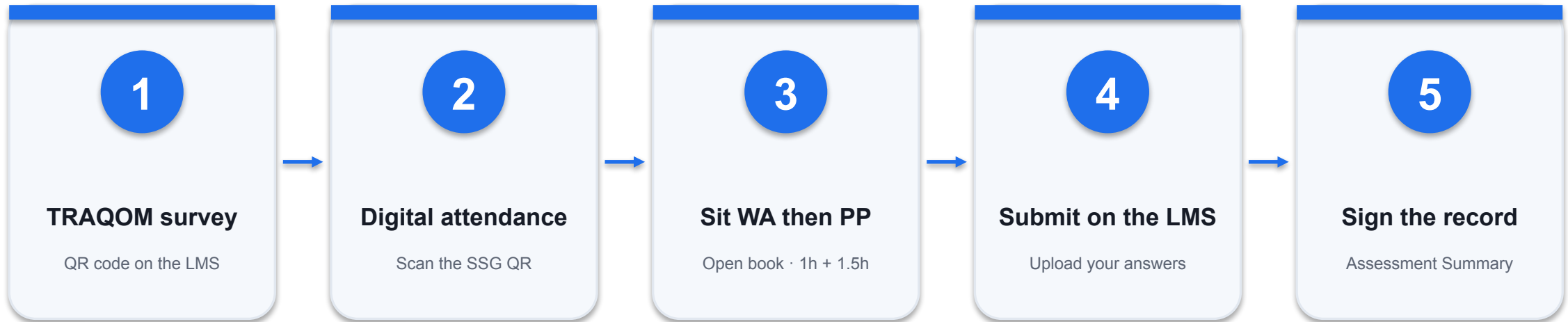
Remember to take the Assessment digital attendance (TRAQOM) before you start.

4

Submit on the LMS

Upload your completed answers at <https://lms-tms.tertiaryinfotech.com/>

Assessment Flow



REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

HAPPY BUILDING

Thank You!

You can now identify integration opportunities, scan them for feasibility, build and test Power Automate flows, create Power Apps, and join the two together into a working, debugged integration.